Darke Theory Field Equation Frozen 3D Parent Action, Post-Threshold Closure Realisation, and Classical Validation

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Abstract

This paper freezes Darke Theory Field Equation at the level of its live classical three-dimensional particle sector and records the executed validation campaign for that frozen object. The aim is exact: to move from conceptual field-theoretic construction to a fixed parent action, a solved topological branch, and a direct classical audit structure capable of surviving basis enlargement. In the broader Darke Theory architecture, post-threshold geometry is the regime in which closure becomes geometrically encoded, and the Higgs is already placed as the first scalar geometric expression of closure. The present paper does not claim a final particle identification. It does claim that DTFE may be the first frozen field-theoretic route by which that post-threshold closure structure begins to be realised explicitly.

The frozen object is a twist-stabilised $SU(2)$ parent theory with left current $L_\mu = U^\dagger \partial_\mu U$ and action

$$S_{\mathrm{DTFE}}[U] = \int d^4x \left[-\frac{1}{2}\,\mathrm{Tr}(L_\mu L^\mu) + \frac{1}{16}\,\mathrm{Tr}\big([L_\mu, L_\nu][L^\mu, L^\nu]\big) \right]. \tag{1}$$

Under the hedgehog ansatz

$$U(\mathbf{x}) = \cos F(R)\,\mathbf{1} + i\,\hat{\mathbf{x}}\cdot\boldsymbol{\tau}\,\sin F(R), \qquad R = |\mathbf{x}|, \tag{2}$$

this reduces to the frozen radial field equation

$$(R^2 + 2\sin^2 F)F'' + 2RF' + \sin(2F)\left(F'^2 - 1 - \frac{\sin^2 F}{R^2}\right) = 0, \tag{3}$$

with

$$F(0) = \pi, \qquad F(\infty) = 0.$$

The corresponding static energy functional is

$$E[F] = 4\pi \int_0^\infty \left[R^2 F'^2 + 2\sin^2 F + 2\sin^2 F\, F'^2 + \frac{\sin^4 F}{R^2} \right] dR. \tag{4}$$

The live branch was solved and then tested through four stacked audit layers: scalarised low-ℓ sectors, enlarged reduced nonradial subspace ladders, a first parent-action tangent-space audit, and an expanded parent-action basis audit. The executed branch gives

$$a \approx 2.007991521992, \qquad E_{\text{dimless}} \approx 145.85782948, \qquad R_{1/2} \approx 0.93647492,$$

$$R_{\text{rms}} \approx 1.42670436, \qquad q_{1/2} \approx 1.82500000.$$

All audited spectra in the executed packages remain strictly positive. In particular, the expanded 72-dimensional parent-action basis audit has lowest eigenvalues beginning at

$$0.1272043,\ 0.1272662,\ 0.1283072,\ 0.1473617,\ \ldots$$

The result established here is concrete and bounded. DTFE is now frozen, executable, and strongly validated at the classical parent-action level within the audited basis framework. That is the strongest honest statement supported by the executed evidence.

1. Introduction

A field equation becomes physically serious when three things are true at once. First, the parent object is fixed rather than continually reinterpreted. Secondly, a live branch exists as an actual solved configuration rather than a symbolic possibility. Thirdly, that branch survives repeated attempts to destabilise it under enlargement of the tested space. This paper records precisely that stage for Darke Theory Field Equation .

The purpose here is narrow and exact. This is not a general philosophical exposition of Darke Theory, and it is not a return to the wider corpus on recursive differentiation, closure, admissibility, and persistent structure. Those belong to the larger programme. The task of the present paper is more specific: to freeze DTFE as a full three-dimensional parent action, isolate its live particle branch, and document the executed classical validation campaign performed directly on that frozen object.

That distinction matters because earlier stages of the programme were necessarily weaker. The route passed through scalar reductions, reduced sectors, and partial effective models. Those stages were useful because they separated trivial survivals from real ones. Some branches failed when the tested structure was too weak to carry the intended content. Others survived but remained reduced surrogates rather than full parent-action theories. The decisive transition occurred once the live branch was frozen as a twist-stabilised three-dimensional hedgehog sector generated by a full $SU(2)$ action rather than by a radial equation standing on its own.

Once that transition was made, the problem changed character. The task was no longer to propose a plausible branch. It was to determine whether a fixed parent object with a

nontrivial topological sector could survive a substantial classical audit stack under direct basis enlargement.

The answer established here is yes, within a clearly bounded sense. DTFE is now a frozen parent-action theory with a solved charge-one branch whose observables remain stable across the executed packages and whose tested fluctuation sectors remain positive through a serious sequence of audits, culminating in a direct 72-dimensional parent-action basis construction. That is already a material shift in status. It means the theory has crossed from conceptual field-theory aspiration into the regime of fixed object, solved branch, and executed classical validation.

There is, however, a second reason this stage matters, and it is the deeper point of the paper. In the broader Darke corpus, post-threshold geometry is not an arbitrary descriptive layer. It is the regime in which closure becomes geometrically encoded. Within that same architecture, the Higgs is not treated as an arbitrary scalar inserted by hand, but as the first scalar geometric expression of closure: the first scalar layer required to encode closure anisotropy once thresholded geometric legibility has been crossed.

That broader claim does not by itself produce a field object. It defines an architectural role. What the present paper provides is more concrete. DTFE is a frozen nonlinear parent-action theory with a live topological branch that survives direct classical audit. In that sense, DTFE may be the first frozen field-theoretic route by which post-threshold closure structure begins to be realised explicitly rather than merely located conceptually. The present paper does not claim that the final particle identification step has already been closed. It does claim that the first serious executable field-theoretic route into that regime now exists.

The paper is organised accordingly. Section 2 states the exact scope of the claim. Section 3 presents the unified field-equation block. Section 4 gives the frozen parent action in full. Section 5 derives the live hedgehog branch. Section 6 records the solved branch observables. Section 7 presents the audit ladder. Section 8 explains where DTFE sits in the broader closure architecture. Section 9 states the physical meaning of the result at the classical level. Section 10 marks the boundary of the present claim. Section 11 closes with the proper validation verdict.

2. Scope and Claim Boundary

The result established in this paper is strong, but it is not unbounded.

The paper does *not* claim that all of Darke Theory has been proved in every regime. It does *not* claim that the full coupled infinite-dimensional nonradial fluctuation problem has been analytically exhausted. It does *not* claim quantum completion, complete phenomenological calibration, or final particle identification.

What it *does* claim is exact:

> There now exists a frozen three-dimensional parent action whose live topological particle branch survives an extended executed classical validation campaign at the parent-action level.

That wording is deliberate. The theory is no longer merely a candidate radial branch, no longer merely a reduced-sector surrogate, and no longer dependent on an unfrozen interpretive object. It is a fixed parent-action theory with a live branch that has survived nontrivial numerical stress under repeated enlargement of the tested space.

The right summary statement is therefore the following:

> Darke Theory Field Equation passes the extended classical validation campaign within the audited parent-action basis framework.

This is the strongest honest statement supported by the executed evidence, and it is the statement the rest of the paper is written to justify.

3. Unified Field Equation Block

For clarity, the full parent object and its live reduced branch are first stated in one place.

The frozen parent field is an $SU(2)$ field

$$U(x) \in SU(2), \qquad L_\mu := U^\dagger \partial_\mu U.$$

The parent action is

$$S_{\mathrm{DTFE}}[U] = \int d^4x \left[-\frac{1}{2}\,\mathrm{Tr}(L_\mu L^\mu) + \frac{1}{16}\,\mathrm{Tr}\Big([L_\mu, L_\nu][L^\mu, L^\nu]\Big) \right]. \tag{5}$$

Its Euler–Lagrange field equation is

$$\partial_\mu \Big(L^\mu - \frac{1}{4}\,[L_\nu, [L^\mu, L^\nu]] \Big) = 0. \tag{6}$$

The live topological particle sector is exposed by the hedgehog ansatz

$$U(\mathbf{x}) = \cos F(R)\,\mathbf{1} + i\,\hat{\mathbf{x}}\cdot\boldsymbol{\tau}\,\sin F(R), \qquad R = |\mathbf{x}|. \tag{7}$$

Under this ansatz the parent theory reduces to the radial field equation

$$\Big(R^2 + 2\sin^2 F\Big)F'' + 2RF' + \sin(2F)\left(F'^2 - 1 - \frac{\sin^2 F}{R^2}\right) = 0, \tag{8}$$

with boundary conditions

$$F(0) = \pi, \qquad F(\infty) = 0. \tag{9}$$

The corresponding static energy functional is

$$E[F] = 4\pi \int_0^\infty \left[R^2 F'^2 + 2\sin^2 F + 2\sin^2 F\, F'^2 + \frac{\sin^4 F}{R^2} \right] dR. \tag{10}$$

This unified block is not a substitute for the fuller separated development below. Its role is to make the full field-theoretic object and its live reduced branch immediately legible in one continuous presentation.

4. Frozen Parent Action

The frozen parent object is the $SU(2)$ theory already stated in (5)–(6). At the level of the present paper, this is the primary object of the theory. It is not a provisional guide to a later effective branch. It is not a symbolic parent waiting to be replaced. It is the fixed object on which the classical validation campaign is performed.

The corresponding static energy is

$$E[U] = \int_{\mathbb{R}^3} d^3x \left[-\frac{1}{2}\,\mathrm{Tr}(L_i L_i) - \frac{1}{16}\,\mathrm{Tr}\Big([L_i, L_j][L_i, L_j]\Big) \right]. \tag{11}$$

The conceptual significance of this section is simple. The radial equation used in the live branch is no longer the starting point. It is a reduction of a fixed three-dimensional parent-action theory. That is precisely what makes the later parent-action audits decisive rather than decorative.

5. Hedgehog Reduction and the Live Branch

To expose the live particle sector, impose again the hedgehog ansatz (7). Under this ansatz, the parent static energy (11) reduces exactly to

$$E[F] = 4\pi \int_0^\infty \left[R^2 F'^2 + 2\sin^2 F + 2\sin^2 F\, F'^2 + \frac{\sin^4 F}{R^2} \right] dR. \tag{12}$$

The corresponding Euler–Lagrange equation is

$$\Big(R^2 + 2\sin^2 F\Big) F'' + 2RF' + \sin(2F) \left(F'^2 - 1 - \frac{\sin^2 F}{R^2} \right) = 0. \tag{13}$$

The branch is fixed by the boundary conditions

$$F(0) = \pi, \qquad F(\infty) = 0, \tag{14}$$

together with the near-origin expansion

$$F(R) = \pi - aR + O(R^3), \qquad R \to 0. \tag{15}$$

The executed live solution is determined by the shooting value

$$a \approx 2.007991521992. \tag{16}$$

Its topological charge is

$$B = -\frac{1}{24\pi^2} \int_{\mathbb{R}^3} \epsilon_{ijk} \operatorname{Tr}(L_i L_j L_k)\, d^3x = 1. \tag{17}$$

Thus the branch is not merely finite-energy and nontrivial in shape. It is a genuine topological particle branch of the frozen parent theory.

At large radius the profile obeys

$$F(R) \sim \frac{C}{R^2}, \qquad R \to \infty, \tag{18}$$

with numerically determined C of order unity. This decay structure is already physically important. It is one of the ingredients that later yields the correct particle-like weak-field behaviour rather than the wrong asymptotic signature of a wall-type object.

6. Solved Branch Observables

The executed branch observables are

$$E_{\text{dimless}} \approx 145.85782948, \qquad R_{1/2} \approx 0.93647492,$$

$$R_{\text{rms}} \approx 1.42670436, \qquad q_{1/2} \approx 1.82500000, \qquad F(q_{1/2}) \approx 0.49991220.$$

These numbers matter for more than record-keeping. They show that the branch is stable as a solved numerical object across the executed packages. The live sector is not being held together by one fragile solver instance or by an underresolved reduction. The branch observables remain fixed within the validation pipeline, which is exactly what one wants from a frozen classical object.

It is also useful to separate two quantities that appear in the executed materials. The dimensionless radial branch energy E_{dimless} is the solved value associated with the hedgehog branch itself. The parent-action grid energy appearing in the direct audit package belongs to the finite three-dimensional audit lattice used to construct the tangent-space Hessian. These are related but not identical numerical objects, and their distinction is entirely appropriate within the validation structure.

7. Executed Classical Validation Campaign

The classical validation campaign consists of four stacked layers:

(1) scalarised low-ℓ audit,

(2) reduced nonradial subspace ladder,

(3) first parent-action tangent-space audit,

(4) expanded parent-action basis audit.

The order matters. Each stage strengthens the result in a different way. The early layers show that there is no easy instability hiding in the simplest angular sectors. The reduced ladder then tests enlargement of the nonradial space. The parent-action audits finally move the argument back onto the frozen three-dimensional object itself.

7.1. Scalarised Low-ℓ Audit

The scalarised low-ℓ audit was executed through $\ell = 8$. Every audited spectrum remained strictly positive. Representative lowest modes are

$$\begin{aligned}
\ell = 0: \quad & [2.00235081,\ 2.04217985,\ 2.11851318,\ 2.23044548,\ 2.37727188,\ 2.55847759,\\
& \quad 2.77364994,\ 3.02243840,\ 3.30453414,\ 3.61965879,\ 3.96755782,\ 4.34799657],\\
\ell = 4: \quad & [2.10698111,\ 2.21863649,\ 2.36060059,\ 2.53349610,\ 2.73745219,\ 2.97250989,\\
& \quad 3.23870073,\ 3.53607031,\ 3.86468587,\ 4.22463624,\ 4.61602733,\ 5.03897489],\\
\ell = 8: \quad & [2.26171373,\ 2.44288895,\ 2.65133638,\ 2.88979591,\ 3.15905031,\ 3.45941035,\\
& \quad 3.79102203,\ 4.15396165,\ 4.54827225,\ 4.97397992,\ 5.43110181,\ 5.91965041].
\end{aligned} \tag{19}$$

No negative scalarised low-angular mode appears.

That point matters. The simplest place for a hidden classical instability to reveal itself is often in the low-angular structure. It does not.

7.2. Reduced Nonradial Subspace Ladder

The reduced nonradial ladder was executed at three successive basis sizes:

$$(\ell_{\max}, n_{\max}) = (5,4), \qquad (7,5), \qquad (9,6).$$

The smallest audited eigenvalues at those levels are

$$\begin{aligned}
(\ell_{\max}, n_{\max}) = (5,4): \quad & \lambda_{\min} \approx 1.14647005 \times 10^{-3},\\
(\ell_{\max}, n_{\max}) = (7,5): \quad & \lambda_{\min} \approx 7.16988913 \times 10^{-5},\\
(\ell_{\max}, n_{\max}) = (9,6): \quad & \lambda_{\min} \approx 5.91068015 \times 10^{-6}.
\end{aligned} \tag{20}$$

These values are small, but their significance is not raw size. Their significance is that they stay positive under substantial enlargement of the tested reduced basis. Nothing flips sign as the ladder grows from dimensions 20 to 35 to 54. The correct reading is not that the branch sits on an obviously massive stability gap in every direction, but that enlargement fails to uncover a negative mode in the tested reduced nonradial space.

7.3. First Parent-Action Tangent-Space Audit

The first direct parent-action tangent-space audit was built from a 9-dimensional $\ell = 1$ basis derived from the frozen three-dimensional parent action itself. The lowest audited eigenvalues are

$$\begin{aligned}[&8.86432813,\ 14.06478118,\ 15.17347222,\ 15.18764552,\ 29.28693777,\\ &29.42713760,\ 29.77217116,\ 30.31548676,\ 30.34379356].\end{aligned} \tag{21}$$

Every tested direction in that first parent-action basis is strongly positive.

This stage is important because the validation object has now shifted back onto the frozen parent theory itself. The argument is no longer being carried only by radial or reduced surrogate sectors. Direct tangent directions of the parent-action object have been tested, and they remain positive.

7.4. Expanded Parent-Action Basis Audit

The expanded parent-action basis audit used

$$\ell = 1, 2, \qquad m = -\ell, \ldots, \ell, \qquad \text{three tangent axes}, \qquad \alpha \in \{0.5, 0.7, 1.0\},$$

for a total basis dimension of 72.

The lowest audited eigenvalues are

$$\begin{aligned}[&0.12720430,\ 0.12726622,\ 0.12830719,\ 0.14736166,\ 0.15058929,\ 0.15109879,\\ &0.15265863,\ 0.15524949,\ 0.15542994,\ 0.63844629,\ 0.65730994,\ 0.66190142,\\ &0.80799231,\ 0.81149455,\ 0.81173280,\ 0.83704328,\ 0.83845803,\ 0.85264549,\\ &0.86712749,\ 0.86734579,\ 0.88375016,\ 0.89718189,\ 0.91019203,\ 0.91074005,\\ &3.76156331,\ 3.88339981,\ 3.91118775,\ 3.96264448,\ 4.93865526,\ 4.98825109].\end{aligned} \tag{22}$$

Again, every listed value is positive.

This is the strongest classical audit layer in the present paper. It is not merely an enlarged reduced surrogate ladder. It is a substantial direct parent-action tangent-space construction. That distinction is what justifies the stronger status now assigned to the theory.

7.5. Validation Verdict

The executed packages produce the verdict ladder

$$\texttt{overall_reduced_classical_pass = True}, \tag{23}$$

$$\texttt{overall_v2_pass = True}. \tag{24}$$

Together with the strictly positive parent-action basis audits, the correct summary verdict is:

> DTFE passes the executed classical validation campaign within the audited parent-action framework.

That is the sentence the audit stack earns.

8. DTFE as an Explicit Post-Threshold Closure Sector

The present result matters not only because a branch survives. It matters because of what kind of branch it is and where it appears in the wider structure of the programme.

In the broader Darke corpus, post-threshold geometry is not primitive. It is the regime in which closure becomes geometrically encoded. Within that same architecture, the Higgs is not treated as an arbitrary scalar inserted by hand, but as the first scalar geometric expression of closure: the first geometric layer required to encode closure anisotropy once the threshold into geometric legibility has been crossed.

That larger claim does not by itself produce a field object. It defines an architectural role. What the present paper provides is more concrete. DTFE is a frozen nonlinear parent-action theory with a live topological branch that survives direct classical audit. In that sense, DTFE may be the first frozen field-theoretic route by which post-threshold closure structure begins to be realised explicitly rather than merely positioned conceptually.

This is the correct strength of the claim at the present stage.

The paper does not claim that DTFE has already completed the final identification with the Higgs boson. That would require more than the present work has earned. It would require additional phenomenological, spectral, and possibly quantum closure. But the architectural consequence is still important: the equation no longer sits outside the corpus as an isolated nonlinear curiosity. It now sits exactly where one would expect the first serious executable field-theoretic realisation of closure-bearing post-threshold structure to begin appearing.

That is the deeper point of the paper.

9. Physical Interpretation of the Frozen Classical Sector

The result is not merely that a numerical branch exists. The result is that the branch has the minimum classical properties one would actually want from a particle-like field-theory sector.

First, the total energy is finite. This excludes the branch from the class of objects that look topologically interesting in a formal sense but fail to define a physically admissible localized sector.

Secondly, the branch is nontrivial and classically stable throughout the executed audits. This matters because a finite-energy branch with an easy negative mode would not define a robust particle sector. What has been shown here is stronger: the object persists under repeated enlargement of the tested space, including direct tangent-space audits of the frozen parent action.

Thirdly, the branch has the correct particle-like weak-field asymptotic structure. Because the energy density is finite and integrable, the sourced weak-field curvature law takes the form

$$K(R) = K_\infty - \frac{\kappa M_{\text{eff}}}{4\pi R} + O(R^{-3}), \qquad R \to \infty. \tag{25}$$

The decisive point is the asymptotic $1/R$ behaviour. Earlier one-dimensional wall-like defects do not give the correct large-distance law. The frozen three-dimensional particle branch does.

So, at the classical level, the live branch now satisfies the three minimum requirements that matter most:

(1) finite total energy,

(2) nontrivial classical stability throughout the executed audit ladder,

(3) particle-like far-field curvature behaviour.

This is the real physical meaning of the paper. The branch is no longer merely mathematically suggestive. It is now a frozen and strongly tested classical sector of a definite parent theory.

10. What Remains Open

The result is major, but it is not final in every possible sense.

This paper does not claim:

(1) exhaustive diagonalisation of the exact coupled vector-spherical-harmonic nonradial Hessian in all channels;

(2) a full infinite-dimensional theorem-proof of every coupled nonradial direction;

(3) complete phenomenological closure of the baryon-scale calibration problem;

(4) quantum completion, spin/isospin quantisation closure, or final particle identification.

Those are stronger targets than the present paper needs in order to establish its actual result. The right status statement is therefore the following:

> Darke Theory Field Equation is frozen and strongly validated at the classical parent-action level within the executed audited basis framework.

That is not a retreat from the result. It is the correct precision of the result.

11. Conclusion

A field equation becomes real when the parent object is fixed, the live branch is solved, and enlargement of the tested space fails to destabilise it. This paper has established that stage for Darke Theory Field Equation .

The frozen parent action is

$$S_{\mathrm{DTFE}}[U] = \int d^4x \left[-\frac{1}{2}\,\mathrm{Tr}(L_\mu L^\mu) + \frac{1}{16}\,\mathrm{Tr}\Big([L_\mu, L_\nu][L^\mu, L^\nu]\Big) \right], \tag{26}$$

with live hedgehog branch

$$\Big(R^2{+}2\sin^2 F\Big)F''{+}2RF'{+}\sin(2F)\left(F'^2 - 1 - \frac{\sin^2 F}{R^2}\right) = 0, \qquad F(0) = \pi, \qquad F(\infty) = 0. \tag{27}$$

The executed observables are

$$a \approx 2.007991521992, \qquad E_{\mathrm{dimless}} \approx 145.85782948, \qquad R_{1/2} \approx 0.93647492,$$

$$R_{\mathrm{rms}} \approx 1.42670436, \qquad q_{1/2} \approx 1.82500000.$$

The branch passes:

- the scalarised low-ℓ audit through $\ell = 8$,
- the enlarged reduced nonradial ladder up to basis dimension 54,
- a first direct parent-action tangent-space audit,
- an expanded 72-dimensional parent-action basis audit.

No negative eigenvalue appears in any executed audited sector.

The correct closing verdict is therefore simple and strong:

> Darke Theory Field Equation is now a frozen, valid, and strongly tested classical field equation within the audited parent-action framework.

That is the strongest honest statement supported by the executed evidence. It also marks a real threshold in the development of the programme: the transition from conceptual field construction to frozen parent-action classical validation, and the first serious executable field-theoretic step toward explicit post-threshold closure realisation.

Appendix: Locked Numerical Outputs

Live Branch

$$a \approx 2.007991521992, \tag{28}$$
$$E_{\text{dimless}} \approx 145.85782948, \tag{29}$$
$$R_{1/2} \approx 0.93647492, \tag{30}$$
$$R_{\text{rms}} \approx 1.42670436, \tag{31}$$
$$q_{1/2} \approx 1.82500000, \tag{32}$$
$$F(q_{1/2}) \approx 0.49991220. \tag{33}$$

Reduced Ladder Minima

$$(\ell_{\max}, n_{\max}) = (5,4): \quad \lambda_{\min} \approx 1.14647005 \times 10^{-3}, \tag{34}$$
$$(\ell_{\max}, n_{\max}) = (7,5): \quad \lambda_{\min} \approx 7.16988913 \times 10^{-5}, \tag{35}$$
$$(\ell_{\max}, n_{\max}) = (9,6): \quad \lambda_{\min} \approx 5.91068015 \times 10^{-6}. \tag{36}$$

Expanded Parent-Action Basis

$$\text{basis dimension} = 72, \tag{37}$$
$$\lambda_{\min} \approx 0.12720430. \tag{38}$$

References

[1] James Alan Darke, *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. `https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry`

[2] James Alan Darke, *2. Darke Theory Ontological Foundation*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. `https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry`

[3] James Alan Darke, *3. Darke Theory Core*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. `https://github.com/DarkeEEandMe/Darke-Theor`

y---Pre-Geometry

[4] James Alan Darke, *Darke Theory: From Recursive Differentiation to Admissibility, Quantum Structure, and Cross-Domain Exclusion*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry

[5] James Alan Darke, *7. The Critical Closure Stability Theorem*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry

[6] James Alan Darke, *5. Recursive Differentiation and the Structural Origin of Quantum Mechanics*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry

[7] James Alan Darke, *Relativistic Closure and the Curvature Bridge*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry

[8] James Alan Darke, *The Coherence Burden Principle*, in *Darke Theory - Pre-Geometry Research Repository*, GitHub repository. https://github.com/DarkeEEandMe/Darke-Theory---Pre-Geometry

[9] T. H. R. Skyrme, *A Non-Linear Field Theory*, *Proceedings of the Royal Society of London A*, **260** (1961), 127–138.

[10] T. H. R. Skyrme, *A Unified Field Theory of Mesons and Baryons*, *Nuclear Physics*, **31** (1962), 556–569.

[11] Edward Witten, *Current Algebra, Baryons, and Quark Confinement*, *Nuclear Physics B*, **223** (1983), 433–444.

[12] P. W. Higgs, *Broken Symmetries and the Masses of Gauge Bosons*, *Physical Review Letters*, **13** (1964), 508–509.

[13] F. Englert and R. Brout, *Broken Symmetry and the Mass of Gauge Vector Mesons*, *Physical Review Letters*, **13** (1964), 321–323.

[14] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble, *Global Conservation Laws and Massless Particles*, *Physical Review Letters*, **13** (1964), 585–587.